

Lightchain P2P

BETA Release Readiness Report

Evidence base for tagging v0.9.0-beta.1

20 August 2026

Repository HEAD: d7cebe4 (main)

Executive summary

The lightchain-p2p desktop application — peer-to-peer chat rooms, a multi-chain wallet, Uniswap swaps, an Ethereum-to-Lightchain bridge, and paid AI inference settled on Lightchain mainnet — has completed a four-sprint remediation and verification program. Every release blocker identified in the audit is closed, the money loop has been exercised end-to-end with real funds on mainnet, and the adversarial suites report zero open findings. The engineering verdict is: **ready to tag v0.9.0-beta.1 once four owner-side items complete** (key ceremony, signing certificates, two seeders, tag push). None of the four requires further code.

Gate	State	Evidence
Unit / integration tests	1,433 passing, 4 skipped (intentional)	<code>pnpm -r test</code> at HEAD d7cebe4
Lint / typecheck / build	Clean across 19 packages	<code>pnpm -r lint / typecheck / build</code>
Live harnesses	259 assertions, 0 failures	<code>docs/audit/2026-08-20-sprint3/</code>
Adversarial	137 probes, 0 findings	<code>adversarial.md</code> (hostile-renderer 118, WSL 19)
Funded mainnet QA	14 / 14 actionable checks	<code>jobs 2756-2757, bridge tx 0x7affc7b6...</code>
CI	harness job blocking; unsigned installers fail the build	<code>.github/workflows/ci.yml</code> , <code>build-matrix.yml</code>

What the program is

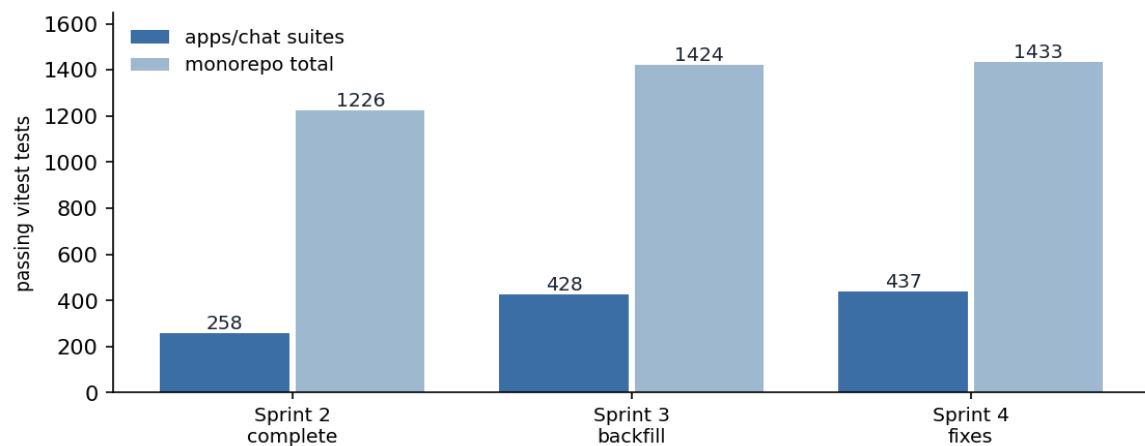
Three processes: an Electron shell, a Bare worker that owns all keys, storage, and networking, and a sandboxed renderer that is only a view. Money moves pass a fail-closed confirmation guard the renderer cannot configure off. Distribution and updates travel over the Holepunch peer network; inference is paid per job into an on-chain session on Lightchain (chain id 9200), with worker answers committed on-chain and evidence sealed locally for disputes.

Sprint results

Sprint	Scope	Outcome
1 — Money safety	The eight audit blockers: chain-aware confirm thresholds, conversation wedge fix, ask limits and fee floor, sealed worker keystore password, stake probe surfacing, chain-aware ledger, swap/bridge allowance fixes, IPC timeouts, trust disclosure, diagnostics export.	All landed; suites green.

Sprint	Scope	Outcome
2 — Recovery	Job lifecycle with claimTimeout / claimRefund / disputeJob, delegate revocation, stake guard against live on-chain minimum, failover RPC pool, 3-confirmation sends with reorg revalidation, retryable OTA apply, swap requote throttling.	All landed; 1,226 tests at integration.
3 — Hardening	170 backfill tests (planSend / planSwap / bridge routes / four quiet handlers), README + CODEOWNERS, harness CI promoted to blocking, full harness re-runs archived, hostile-renderer and WSL adversarial re-runs, live QA pass.	Zero security findings; two stale probes identified and handed to Sprint 4.
4 — Release execution	Blocking signature gates, tag-triggered release workflow (draft releases), OTA smoke harness, Linux distribution decision (ADR 0007), two-seeder posture, all Sprint 3 findings fixed, release checklist, funded-wallet harness, key-ceremony runbook.	Complete; tree clean at d7cebe4.

Test progression



The chat application suite grew 258 to 437 tests across Sprints 3-4, all of the growth aimed at money-adjacent planners and previously uncovered handlers. The monorepo total stands at 1,433.

Live mainnet verification, 20 August 2026

The funded-wallet harness drove the production build against Lightchain mainnet with the owner's real wallet (0xB00561fe...Efd7fc21, funded with 5,474.98 LCAI). Confirmation dialogs were approved by the owner in the window; the harness never answers the guard itself.

Flow	Result	On-chain evidence
Prepaid inference	24.8 LCAI prepaid, delegate authorized	delegate status read live
Paid ask, job 2756	answered, settled, 0.02 LCAI fee	worker 0xFCB6b564...0501D9
Paid ask, job 2757	answered, settled, 0.02 LCAI fee	worker 0xdB258B4b...A48a16E2Cf
Commitments + evidence	commitment push followed both answers; sealed evidence stored for both jobs	hasEvidence: true
Dispute window	tracked, open until 2026-08-22; filing skipped (risks bond on a correct answer)	ai.jobState
Bridge, Lightchain to Ethereum	1-wei smoke transfer confirmed	tx 0x7affc7b6...b7d2, block 1548073, dispatch 0xae2de8fda9...
Bridge persistence	pending entry and disclosure survived a full app restart	funded-check --check-pending
Guard fail-closed	unapproved bridge attempt timed out with nothing signed; forged dialog answers settle nothing	locking-check 29/29

Two checks skipped structurally: no job of this wallet has ever timed out unanswered, so there is nothing to claim a timeout or refund against. That path is covered by 19 pinned unit tests instead.

Security posture

The hostile-renderer suite — a compromised window probing every worker handler, including all handlers added in Sprints 1-2 — passed 118/118 with no finding. The WSL adversarial suite passed 19/19 across a real NAT boundary (forged signatures arrive verified=false; no plaintext at rest). The diagnostics export was scanned live for seed phrases, wallet passwords, and the suite's own hostile payloads: none present. Password-gated refusal ordering was re-proven end-to-end after the locking suite was rewritten for the dialog-guard design.

Known limitations, stated plainly

Item	Status
OTA live-apply smoke	Skipped honestly: the dev update channel has never carried a real chat build (it holds the upstream template). Closes when the production link stages a real build with the by-arch layout.
Approved worker registration	Not yet exercised; the machine key holds 50,001 LCAI and the guard, dialog, and pre-Docker ordering are verified. Needs Docker on the host.
Swap dust floor	Applies to ether-denominated inputs only; non-ether tokens share no unit with the fee and stay on balance checks. Documented in code.
Flatpak build	Manifest repaired and self-build documented; not compiled here (no flatpak-builder on Windows). ApplImage is the published Linux artifact; Snap is not shipped (ADR 0007).
Two-instance reconnects	Harness needs two storages; single-instance loopback passed 4/5 (the one fail: no peers online, environmental).

Remaining work — all owner-side, no code

#	Item	Runbook
1	Key ceremony: production multisig pear link, three machines, one sitting	docs/key-ceremony.md
2	Signing certificates (Windows OV / Azure, Apple Developer) — multi-week lead, start immediately	docs/signing-procurement.md
3	Stand up two always-on seeders on independent providers	docs/seeders.md
4	Push tag v0.9.0-beta.1 — the release workflow builds, verifies signatures, and drafts the GitHub Release	docs/release-checklist.md

Until the certificate secrets exist in repository settings, any tag push will fail at the signature gates — that is the intended blocking posture, not a malfunction.